American Computer Science League

2022-2023 ● Contest 2: Short Problems ● Senior Division

1. Prefix/Infix/Postfix Notation

Evaluate the following prefix expression if all numbers are single digits:

+ * 3 − 4 / 6 2 ^ + − 2 * 2 2 6 2

A. 20
B. 19
C. 18
D. 17
E. 16

2. Prefix/Infix/Postfix Notation

Convert the following infix expression to postfix:

$$\sqrt{1 + \sqrt{1 + \sqrt{1 + x}}}$$

A. 11x1+12/^+12/^+12/^
B. 111x12/^++12/^+12/^
C. 111x+12/^+12/^+12/^
D. 1x11+12/^+12/^+12/^
E. x111+12/^+12/^+12/^

3. Bit-String Flicking

Evaluate the following expression:

(RCIRC-2 (LSHIFT-1 ((RSHIFT-1 01110) AND (LCIRC-2 10011)
OR (NOT 01101))))

A. 10010
B. 00111
C. 10110
D. 00011
E. 01100

4. Bit-String Flicking

Solve for X, a 5-bit string:

(LCIRC-2 X) OR (RSHIFT-2 01101) AND (NOT 11011)
= (LSHIFT-2 (RCIRC-2 10110))

A. 00101
B. 00100
C. 00011
D. 00000
E. 10101

5. LISP

Evaluate the following sequence of LISP expressions:

(SETQ M '(1 2 (5 1) (3 (4 5) 6) (2 (4))))
(CDR (CAR (CDR (REVERSE (CDR (CDR M))))))

A. (5 1)
B. (4 5)
C. ((4 5) 6)
D. (3 (4 5))
E. (2 (4))

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1. 前缀/中缀/后缀

如果下方所有数字都是一个单独的数值，请计算下方前缀表达式的值：

+ * 3 − 4 / 6 2 ^ + − 2 * 2 2 6 2

A. 20
B. 19
C. 18
D. 17
E. 16

2. 前缀/中缀/后缀

请将下方中缀表达式转换为后缀表达式：

$$\sqrt{1 + \sqrt{1 + \sqrt{1 + x}}}$$

A. 11x1+12/^+12/^+12/^
B. 111x12/^++12/^+12/^
C. 111x+12/^+12/^+12/^
D. 1x11+12/^+12/^+12/^
E. x111+12/^+12/^+12/^

3. 比特位操作

计算下列表达式：

(RCIRC-2 (LSHIFT-1 ((RSHIFT-1 01110) AND (LCIRC-2 10011)
OR (NOT 01101))))

A. 10010
B. 00111
C. 10110
D. 00011
E. 01100

4. 比特位操作

X 是一个 5 位的位串，求 X。

(LCIRC-2 X) OR (RSHIFT-2 01101) AND (NOT 11011)
= (LSHIFT-2 (RCIRC-2 10110))

A. 00101
B. 00100
C. 00011
D. 00000
E. 10101

5. LISP 语言

计算下列 LISP 表达式：

(SETQ M '(1 2 (5 1) (3 (4 5) 6) (2 (4))))
(CDR (CAR (CDR (REVERSE (CDR (CDR M))))))

A. (5 1)
B. (4 5)
C. ((4 5) 6)
D. (3 (4 5))
E. (2 (4))